ARRAYS

Reverse an array

Enabling `extdebug` allows access to the `BASH_ARGV` array which stores the current function's arguments in reverse.

CAVEAT: Requires `shopt -s compat44` in bash 5.0+.

Example Function:

```
reverse_array() {
    # Usage: reverse_array "array"
    shopt -s extdebug
    f()(printf '%s\n' "${BASH_ARGV[@]}"); f "$@"
    shopt -u extdebug
}
```

Example Usage:

```
$ reverse_array 1 2 3 4 5
5
4
3
2
1

$ arr=(red blue green)
$ reverse_array "${arr[@]}"
green
blue
red
```

Remove duplicate array elements

Create a temporary associative array. When setting associative array values and a duplicate assignment occurs, bash overwrites the key. This allows us to effectively remove array duplicates.

CAVEAT: Requires bash 4+

CAVEAT: List order may not stay the same.

Example Function:

```
remove_array_dups() {
    # Usage: remove_array_dups "array"
    declare -A tmp_array

    for i in "$@"; do
        [[ $i ]] && IFS=" " tmp_array["${i:- }"]=1
    done
```

```
    printf '%s\n' "${!tmp_array[@]}"
}
```

Example Usage:

```
$ remove_array_dups 1 1 2 2 3 3 3 3 3 4 4 4 4 4 5 5 5 5 5 5
1
2
3
4
5

$ arr=(red red green blue blue)
$ remove_array_dups "${arr[@]}"
red
green
blue
```

Random array element

Example Function:

```
random_array_element() {
    # Usage: random_array_element "array"
    local arr=("$@")
    printf '%s\n' "${arr[RANDOM % $#]}"
}
```

Example Usage:

```
$ array=(red green blue yellow brown)
$ random_array_element "${array[@]}"
yellow

# Multiple arguments can also be passed.
$ random_array_element 1 2 3 4 5 6 7
3
```

Cycle through an array

Each time the `printf` is called, the next array element is printed. When the print hits the last array element it starts from the first element again.

```
arr=(a b c d)

cycle() {
    printf '%s ' "${arr[${i:=0}]}"
    ((i=i>=${#arr[@]}-1?0:++i))
}
```

Toggle between two values

This works the same as above, this is just a different use case.

```
arr=(true false)

cycle() {
    printf '%s ' "${arr[${i:=0}]}"
    ((i=i>=${#arr[@]}-1?0:++i))
}
```